Electronics Research Institute

Register-Transfer Level

Presented to Dr. Heba Draz, ECIP Internship Program Committee
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Course: Hardware/Software Co-Design
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https://eri.sci.eg

FPGA Flow: Register-Transfer Level

- SystemVerilog to Synthesizable Verilog

Mk mode → goldmodel python

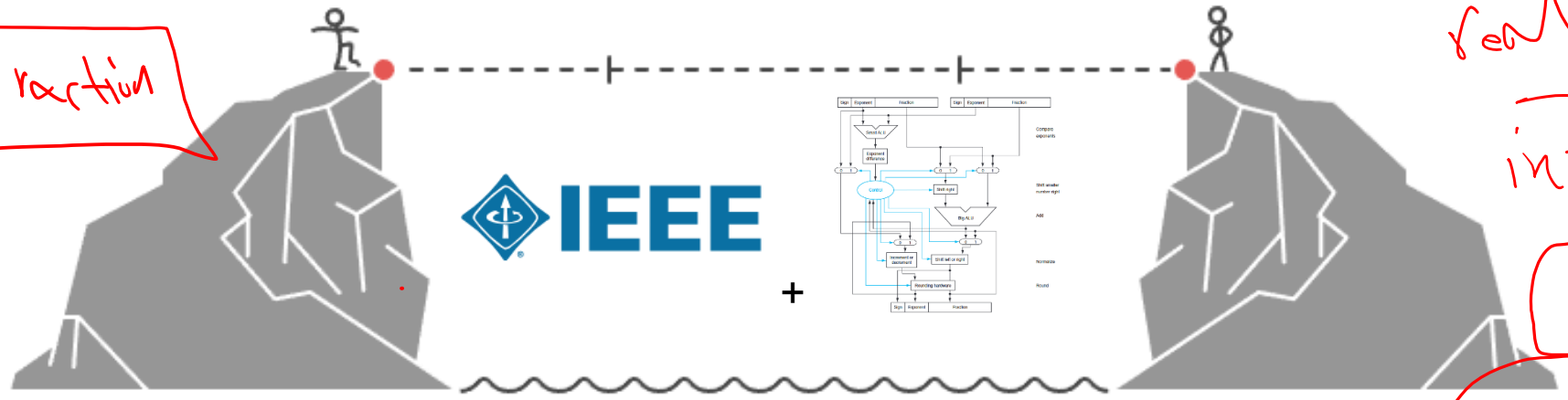
SystemVerilog Design
Non-synthesizable RTL code

IEEE 754 Conversion
Implement floating-point format

Floating-Point Arithmetic
Add arithmetic unit

Synthesizable Verilog
Hardware-ready RTL code

Sign | ex | fraction
1



real → int → OUT

3.14

→ 254.. code

- https://github.com/dawsonjon/fpu/tree/master/float_to_int
- https://github.com/akilm/FPU-IEEE-754/tree/main

FPGA Flow: Register-Transfer Level

- Design Modules

1000

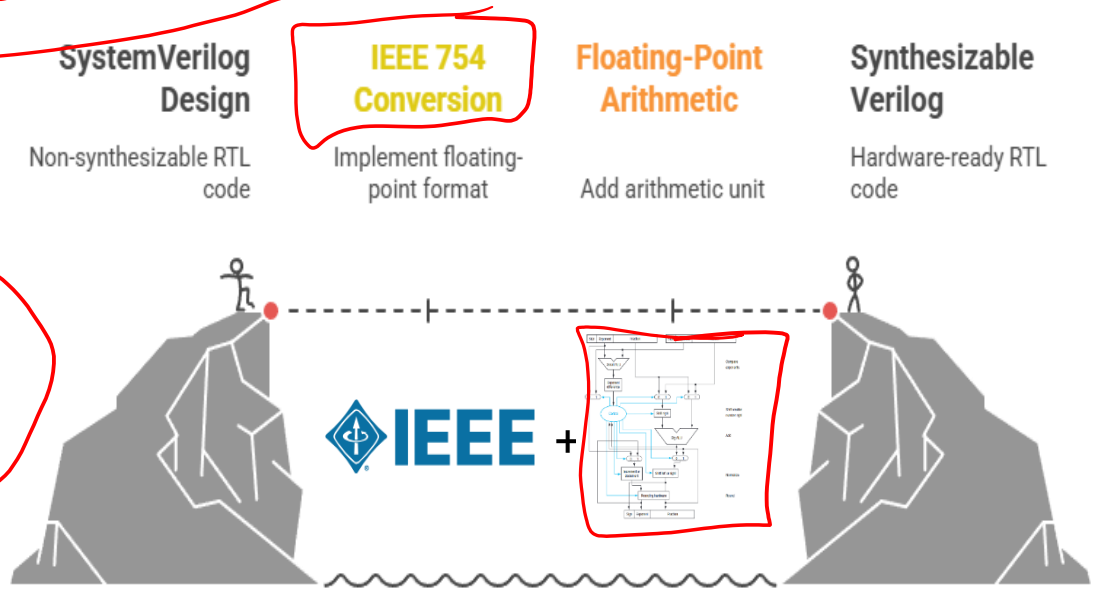
10

Design

- fpu.v: int_to_float, float_to_int.v
- FloatingSqrt.v
- FloatingAddition.v
- FloatingCompare.v
- FloatingDivision.v
- FloatingMultiplication.v

IEEE Standard

FP Arith



FPGA Flow: Register-Transfer Level

- Design Flow

Code

Inputs → IEEE Conversion Module → Floating Point Arithmetic Unit → Crop Yield Predictor → IEEE Conversion Module → Output

real

6.5

Soil_ph

temp_c

33.5

DUT

yield

(output)

4453.5

Golden

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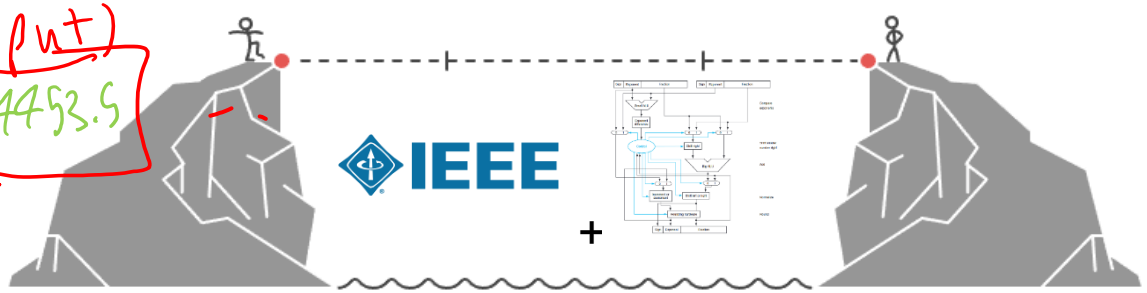
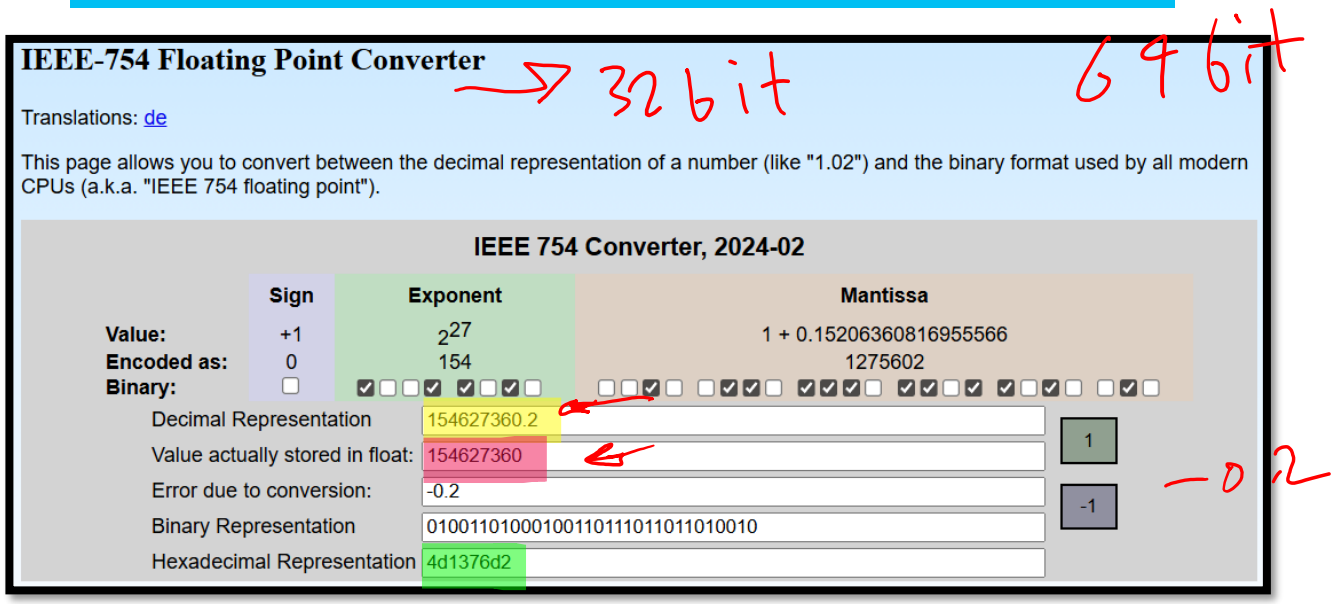


FPGA Flow: Tiny Testbench

- fpu.v: int_to_float, float_to_int.v



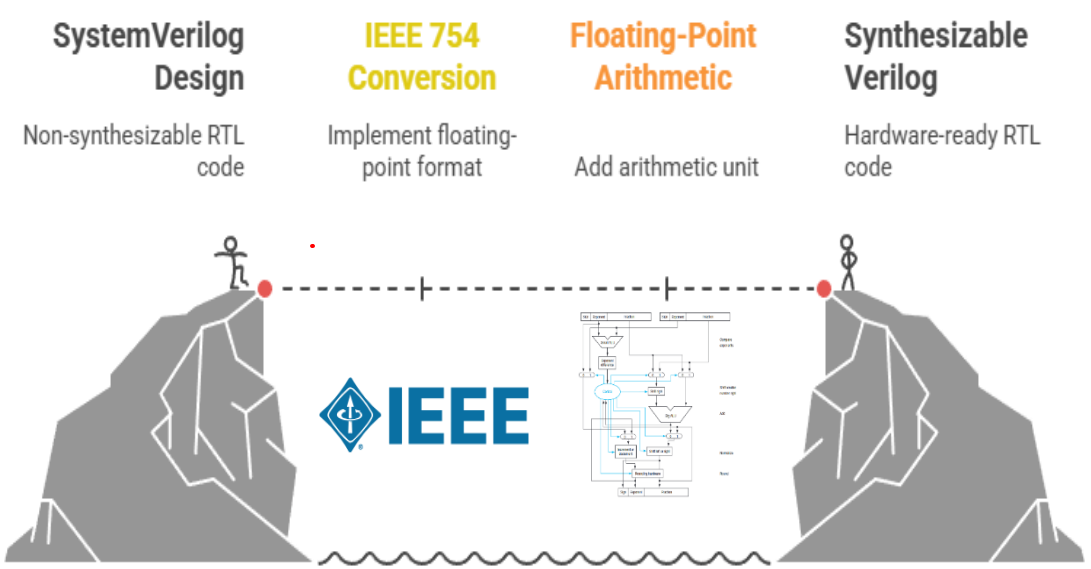


Figure 1: IEEE 754 floating point converter (website: https://www.h-schmidt.net/FloatConverter/IEEE754.html)

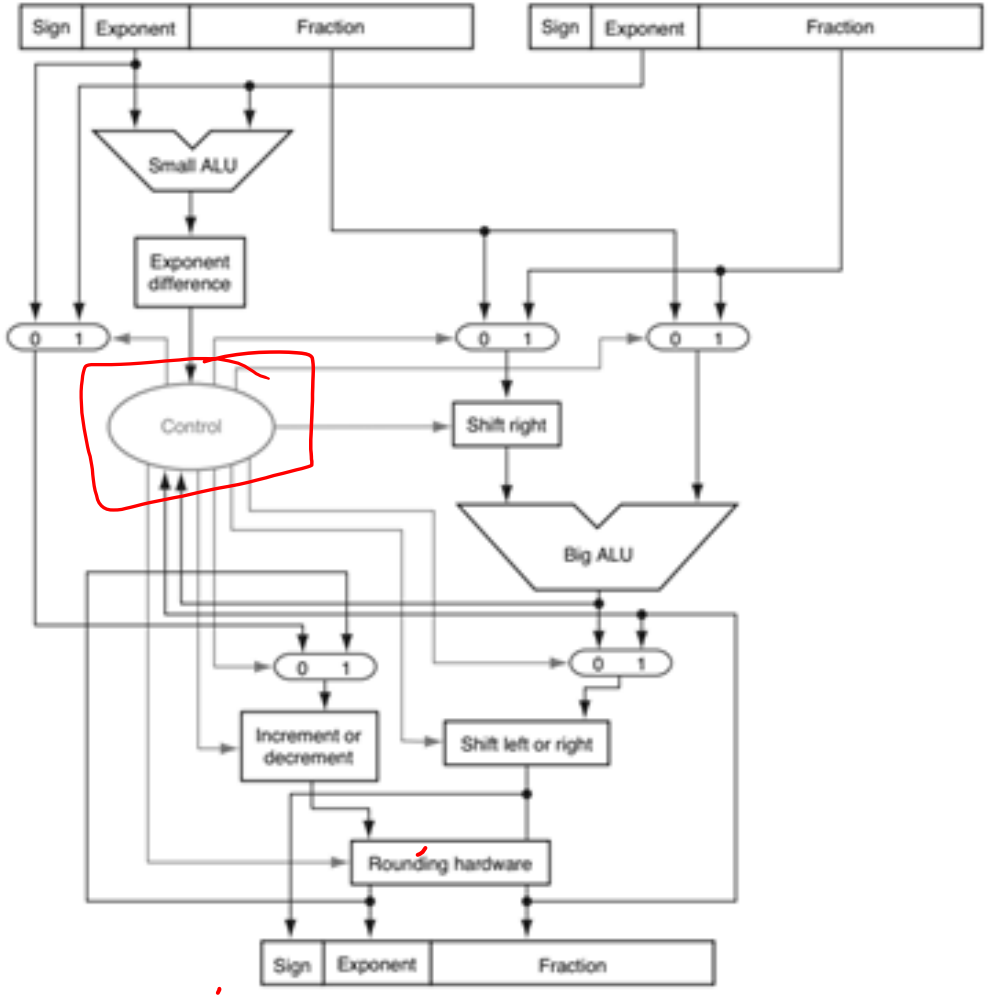
```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS
# PASSED: float=202894552.000000 bits=0x4d417ece DUT_fpu=202894560 REF=202894560
# PASSED: float=154627360.200000 bits=0x4d1376d2 DUT_fpu=154627360 REF=154627360
# PASSED: float=40591483.000000 bits=0x4c1ad81f DUT_fpu=40591484 REF=40591484
# PASSED: float=185025246.400000 bits=0x4d30742e DUT_fpu=185025248 REF=185025248
```

Figure 2: Tiny testbench output, DUT: fpu.v

Questions

- Block diagram of an arithmetic unit dedicated to floating-point addition

paper →



Questions

- Kaggle Dataset

datasets > Smart_Farming_Crop_Yield_2024.csv > data

```
farm_id,region,crop_type,soil_moisture_%,soil_pH,temperature_C,
FARM0001,North India,Wheat,35.95,5.99,17.79,75.62,77.03,7.27,No
FARM0002,South USA,Soybean,19.74,7.24,30.18,89.91,61.13,5.67,Sp
FARM0003,South USA,Wheat,29.32,7.16,27.37,265.43,68.87,8.23,Dri
FARM0004,Central USA,Maize,17.33,6.03,33.73,212.01,70.46,5.03,S
FARM0005,Central USA,Cotton,19.37,5.92,33.86,269.09,55.73,7.93,
FARM0006,Central USA,Rice,44.91,5.78,24.87,238.95,83.06,4.92,Sp
FARM0007,North India,Soybean,36.28,7.04,21.8,123.38,47.91,4.02,
FARM0008,East Africa,Maize,27.1,5.72,22.26,296.33,80.34,5.44,Sp
FARM0009,Central USA,Soybean,40.54,6.35,19.24,184.82,76.5,5.21,
FARM0010,East Africa,Rice,10.25,6.92,16.18,66.85,41.57,5.98,Spr
FARM0011,South India,Wheat,13.39,6.36,23.47,166.76,76.45,8.04,D
FARM0012,South USA,Rice,40.61,5.65,31.16,263.99,44.9,7.91,Drip,
```

1 crop ✓

yield

target

crop-type

ABC

Thank you

Any Questions?